



## Python In Excel: How About a BYOP Option?

Python in Excel is pretty amazing. Imagine if it had Bring Your Own Python support!

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# Python in Excel

## The Good

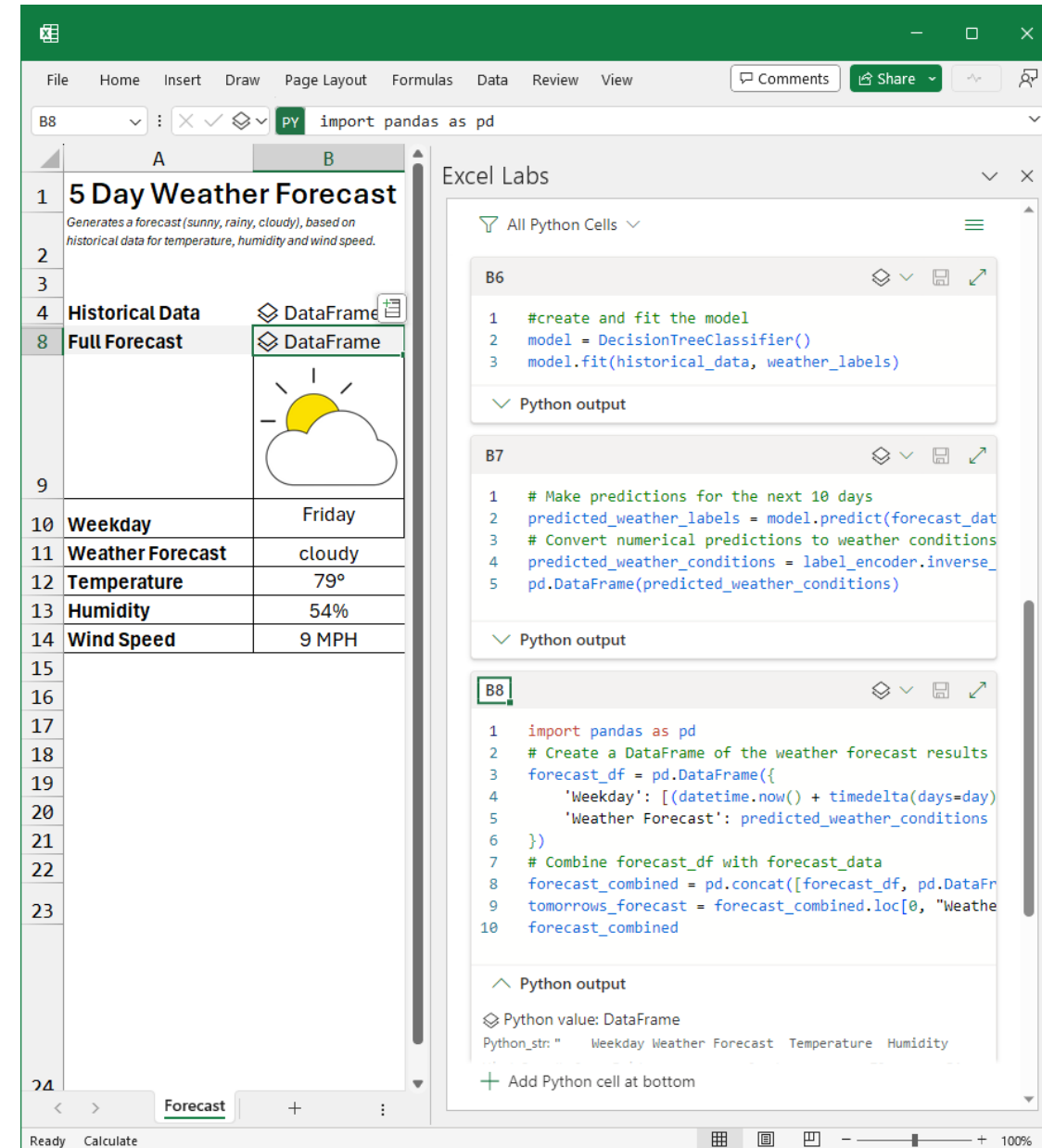
- Solid Microsoft/Anaconda product, not a VBA hack ☺
- Free basic access to secure Python instance run on secure Azure sandbox
- Robust In-Excel Python editor
- Seamless transfer of data/images
- Copilot & advanced compute support (plan upgrades)

## The Bad

- Enterprise-level security, but at what cost in functionality?
- Rate-limited (compute) and feature-limited (Copilot) features may require additional spend
- Python in-Excel Objects and editor not exposed to VBA

## What's Needed

- Bring-Your-Own-Python (BYOP)!
- Allow users to point to local or remote Python instances of their choice
- Expose Python-in-Excel objects and editor to end users



The screenshot displays the Microsoft Excel interface with a Python script running in the background. The script is titled "5 Day Weather Forecast" and generates a forecast based on historical data for temperature, humidity, and wind speed. The output is shown in the Excel grid, with a forecast for Friday, cloudy weather, 79°F temperature, 54% humidity, and 9 MPH wind speed. The Python editor on the right shows the code for creating the model, making predictions, and combining the forecast with historical data.

```
import pandas as pd
```

```
#create and fit the model
model = DecisionTreeClassifier()
model.fit(historical_data, weather_labels)
```

```
# Make predictions for the next 10 days
predicted_weather_labels = model.predict(forecast_data)
# Convert numerical predictions to weather conditions
predicted_weather_conditions = label_encoder.inverse_transform(predicted_weather_labels)
pd.DataFrame(predicted_weather_conditions)
```

```
import pandas as pd
# Create a DataFrame of the weather forecast results
forecast_df = pd.DataFrame({
    'Weekday': [(datetime.now() + timedelta(days=day)).strftime('%A') for day in range(5)],
    'Weather Forecast': predicted_weather_conditions
})
# Combine forecast_df with forecast_data
forecast_combined = pd.concat([forecast_df, pd.DataFrame(forecast_data)], axis=1)
tomorrows_forecast = forecast_combined.loc[0, "Weather Forecast"]
forecast_combined
```

| Weekday | Weather Forecast | Temperature | Humidity | Wind Speed |
|---------|------------------|-------------|----------|------------|
| Friday  | cloudy           | 79°         | 54%      | 9 MPH      |



# Satya Nadella ❤️ Excel!



Microsoft CEO Satya Nadella Giving Excel Demo | 1993



# PyData Boston

DECEMBER 8-10, 2025

TICKETS

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## Keynote Speakers





NEXT STEPS

# A BYOP Option for Python in Excel?

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